

Proton therapy is an important method for treating cancerous tumors by the delivery of radiation. A beam of high energy protons ionizes water molecules, creating free radicals (molecules with one unpaired valence electron) that damage the tumor's DNA and prevent the tumor's growth by eventually leading to cell death.

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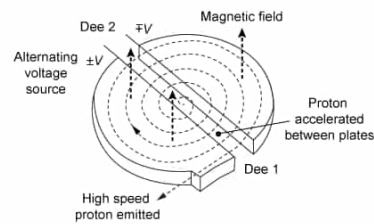


Figure 1 Cyclotron schematic showing the spiraling path of motion (dashed line) traversed by protons

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When the protons enter the body, the stopping power (ie, the effective retarding force on the protons as a function of depth) follows the Bragg curve shown in Figure 2.

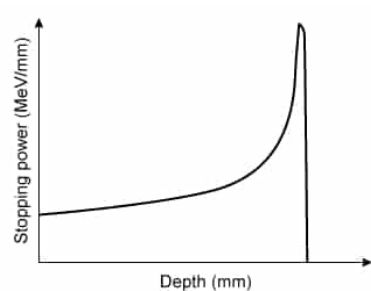
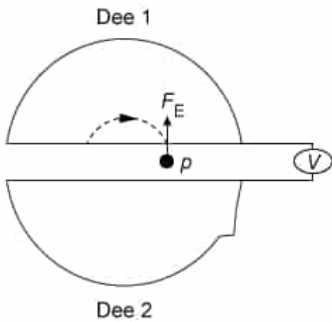


Figure 2 Stopping power vs depth in tissue

Because most of the protons' energy loss is concentrated near the peak of the curve, the radiation can be targeted directly onto the tumor at a specific depth. (Note: $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$)

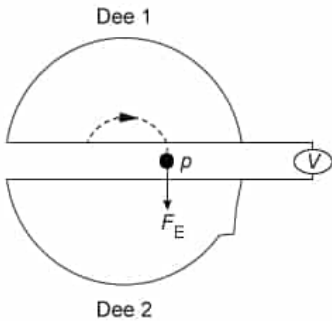
Which image best illustrates the electric force F_E on the proton p as it passes between the dees in the cyclotron?

☐ A.



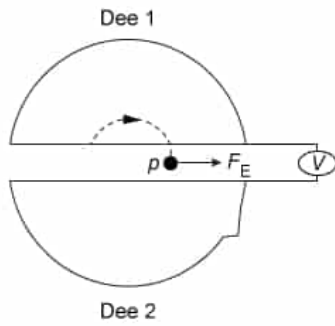
[14%]

☒ B.



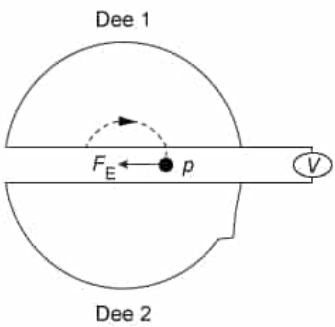
[37%]

☐ C.



[21%]

☐ D.

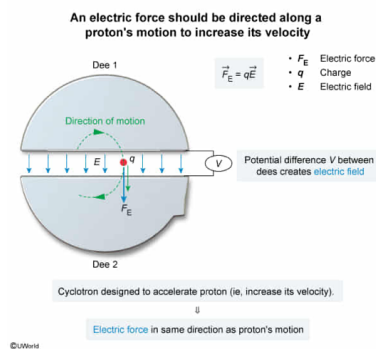


[26%]

Correct

36% answered correctly

Explanation:



The **electric force** F_E on a **charged object** equals the product of the object's charge q and the **electric field** E :

$$\vec{F}_E = q \vec{E}$$

An electric field exists when there is a difference in the **electric potential (ie, voltage)** ΔV between two locations. The **component** of the electric field in a given direction (eg, the y -direction) equals the ΔV per unit of distance Δy :

$$E_y = \frac{\Delta V}{\Delta y}$$

In this question, the potential difference produced by the voltage source in the region between the dees creates an electric field that exerts an F_E on the proton. **Newton's second law of motion** implies that the proton experiences an **acceleration** equal to the ratio of F_E and the proton's mass m :

$$\vec{a} = \frac{\vec{F}_E}{m}$$

Because the cyclotron is designed to increase the proton's velocity over a time interval (ie, create positive acceleration), the direction of the electric force between the dees must be in the same direction as the proton's velocity. Therefore, the electric force exerted on the

proton in the image must be *directed downward* because the proton's velocity is clockwise and downward as it passes between the dees, as shown in Choice B.

(Choice A) In this image, the electric force is opposite the proton's direction of motion. Therefore, the proton will experience negative acceleration, causing it to slow down instead of speeding up.

(Choice C) An electric force directed to the right would cause the proton to accelerate to the right, which is inconsistent with cyclotron design.

(Choice D) This image shows the (leftward) direction of the **magnetic Lorentz force** on the proton. The Lorentz force creates the **centripetal acceleration** that causes the proton to move along a curved path.

Educational objective:

A potential difference creates an electric field that exerts an electric force on charges, causing them to accelerate. An object experiences positive acceleration when a force is exerted in the same direction as the object's motion.

Time spent:0 sec
Subject:
Physics

QID:403114
System:
Mechanics and Energy

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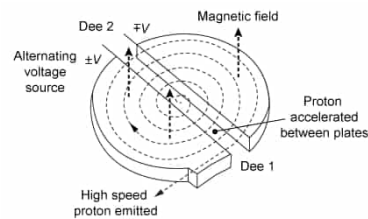


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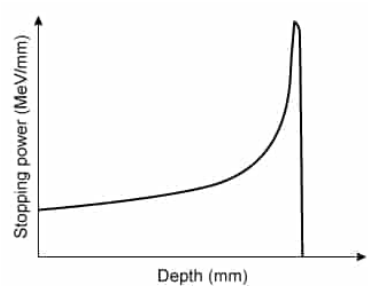


Figure 2 Stopping power vs depth in tissue

Because most of the protons' energy loss is concentrated near the peak of the curve, the radiation can be targeted directly onto the tumor at a specific depth. (Note: $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$)

If the cyclotron producing protons for the beam is replaced with another cyclotron in which the kinetic energy of the emitted protons is doubled, what is the ratio of the speed of the protons produced by the original cyclotron to the speed of protons emitted by the new cyclotron?

- ✔ ☒ A. $1:\sqrt{2}$ [49%]
- ☐ B. 1:2 [16%]
- ☐ C. 1:4 [24%]
- ☐ D. $\sqrt{2}:1$ [9%]

Correct

49% answered correctly

Explanation:

Determining the ratio of proton speeds after doubling kinetic energy

$KE = \frac{1}{2}mv^2$

- KE Kinetic energy
- m Mass
- v Speed

Original cyclotron

New cyclotron (KE doubled)

$$\frac{KE_{original}}{KE_{new}} = \left(\frac{v_{original}}{v_{new}}\right)^2 = \frac{1}{2} \Rightarrow \frac{v_{original}}{v_{new}} = \frac{1}{\sqrt{2}}$$

Ratio of original proton speed to new proton speed is $1:\sqrt{2}$

Kinetic energy *KE* is the energy associated with an object's motion and is equal to half the product of the object's mass *m* and the square of its speed *v*:

$$KE = \frac{1}{2}mv^2$$

In this question, the new cyclotron emits protons with a KE that is double the KE of the original cyclotron:

$$KE_{\text{new}} = 2 \cdot KE_{\text{original}}$$

or, equivalently:

$$KE_{\text{original}} = \frac{KE_{\text{new}}}{2}$$

Rewriting this equation as a ratio and inserting the formula for KE in terms of proton mass m and the original and new proton velocities, v_{original} and v_{new} , yields:

$$\frac{KE_{\text{original}}}{KE_{\text{new}}} = \frac{1}{2}$$

$$\frac{\frac{1}{2}mv_{\text{original}}^2}{\frac{1}{2}mv_{\text{new}}^2} = \frac{1}{2}$$

$$\left(\frac{v_{\text{original}}}{v_{\text{new}}}\right)^2 = \frac{1}{2}$$

Taking the square root of the above equation gives the following ratio:

$$\frac{v_{\text{original}}}{v_{\text{new}}} = \frac{1}{\sqrt{2}}$$

Therefore, the ratio of the original proton speed to the new proton speed is $1:\sqrt{2}$.

(Choice B) 1:2 is the ratio of the squared velocities v^2 and results from neglecting to take the square root in the ratio of the kinetic energies.

(Choice C) 1:4 follows from incorrectly taking the square of the new proton speed.

(Choice D) $\sqrt{2}:1$ is the ratio of the new proton speed to the original proton speed.

Educational objective:

The kinetic energy of an object depends on the object's mass and the square of its velocity. Doubling the kinetic energy increases an object's speed by a factor of $\sqrt{2}$.

Time spent: 0 sec
Subject:
Physics

QID: 403115
System:
Mechanics and Energy

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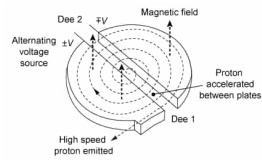


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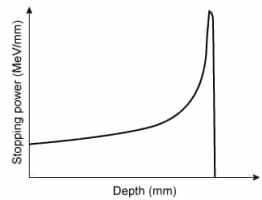


Figure 2 Stopping power vs depth in tissue

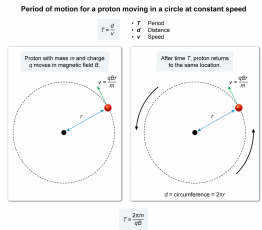
Because most of the protons' energy loss is concentrated near the peak of the curve, the radiation can be targeted directly onto the tumor at a specific depth. (Note: $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$)

A magnetic field B causes a proton of mass m and charge q to move in a circular path of radius r with constant speed $\frac{qBr}{m}$. What is the time required for the proton to complete one cycle of motion around the circle?

- ☐ A. $\frac{m}{qBr}$ [7%]
- ☐ B. $\frac{m}{qB}$ [7%]
- ☐ C. $\frac{qB}{2\pi m}$ [26%]
- ☒ D. $\frac{2\pi m}{qB}$ [58%]

Correct
58% answered correctly

Explanation:



Periodic motion occurs when an object moves cyclically, returning to the same location over and over in a repetitive fashion. The **period** T of a moving object is defined as the time it takes the object to complete one full cycle of motion. Examples of periodic motion include the oscillatory motion of waves and the behavior of a **compressed (or stretched) spring** after it is released.

In this question, the magnetic field B creates a force that causes a proton with charge q to move in a circular path of radius r with a constant speed v :

$$v = \frac{qBr}{m}$$

Consequently, the proton's motion is periodic, and the T needed to complete one cycle of motion around the circle can be determined using the kinematic equation relating v to the ratio of distance traveled d to T :

$$v = \frac{d}{T}$$

Solving this equation for T yields:

$$T = \frac{d}{v}$$

The distance needed to complete one cycle around the circle is equal to the **circumference**, hence:

$$d = 2\pi r$$

Inserting this formula into the equation for T gives:

$$T = \frac{2\pi r}{\frac{qBr}{m}}$$
$$T = (2\pi r) \left(\frac{m}{qBr} \right)$$

Therefore, the proton's period equals:

$$T = \frac{2\pi m}{qB}$$

(Choice A) $\frac{m}{qBr}$ is the inverse of the proton's speed, which is not a meaningful physical quantity.

(Choice B) $\frac{m}{qB}$ results from neglecting the factor of 2π in the equation for circumference.

(Choice C) $\frac{qB}{2\pi m}$ is the frequency of the proton's motion (ie, the number of cycles per unit time), which is the inverse of the period.

Educational objective:

Periodic motion describes an object moving in a cyclical and repetitive fashion. The period of an object moving in a circular path can be determined by dividing the circle's circumference by the object's speed.

Time spent:0 sec
Subject:
Physics

QID:403116
System:
Mechanics and Energy

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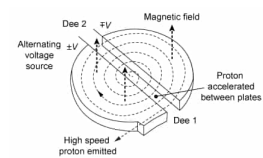


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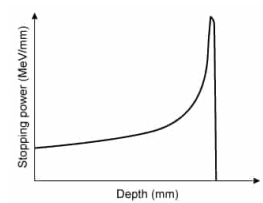


Figure 2 Stopping power vs depth in tissue

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What physical quantity is best represented by the area under the curve in the graph in Figure 2?

- ☐ A. Total distance traveled by the proton [3%]
- ☒ B. The work done on the proton [77%]
- ☐ C. Average force on the proton [14%]
- ☐ D. Average acceleration of the proton [3%]

Correct
78% answered correctly

Explanation:

Area under a Bragg curve equals the work done to stop protons

$W = \text{Area} = F \cdot d$

- W Work done
- F Force
- d Distance

$\text{Area} = \text{Stopping power} \times \text{Depth}$

$\text{Units of area} = \frac{\text{MeV}}{\text{mm}} \times \text{mm} = \text{MeV}$

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The **work** W done on an object is equal to the product of the **component** $F_{\parallel}$ of a constant force exerted along the direction of an object's motion and the distance d that the object moves as the force is exerted:

$$W = F_{\parallel} d$$

The work done has units of **energy** (ie, the product of force and distance) because the work represents a change in the object's energy. When a graph of force as a function of distance is given (ie, the force is not constant), the work done is equal to the product of the vertical and horizontal axis variables (F and d , respectively), which is the **area under the curve**:

$$\text{Area} = W$$

In this question, the vertical and horizontal axes of the graph in Figure 2 are the stopping power and the depth, respectively. The stopping power is a force because it acts to slow the motion of the proton, so it also has units of force (ie, energy per unit of distance, MeV/mm). Consequently, the area under the curve in the graph in Figure 2 is the product of force and distance, and it represents the work done to stop the proton:

$$\text{Area} = \text{force} \times \text{depth}$$

In terms of the units of the variables, which are represented by brackets:

$$\begin{aligned} \left[\text{Area} \right] &= \frac{\text{MeV}}{\text{mm}} \times \text{mm} \\ \left[\text{Area} \right] &= \text{MeV} = \left[W \right] \end{aligned}$$

Therefore, the area under the curve best represents the work done on the proton.

(Choice A) The distance traveled by the proton equals the area under the curve in a graph of proton *velocity as a function of time*.

(Choices C and D) The area under the curve has units of energy, and therefore cannot represent the average force or the average **acceleration**.

Educational objective:

A force has units of energy per unit distance. The area under the curve in a graph of force versus distance equals the work done on an object.

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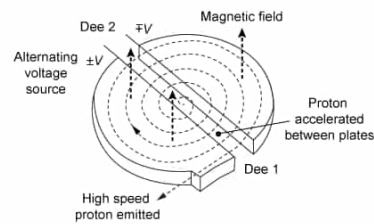


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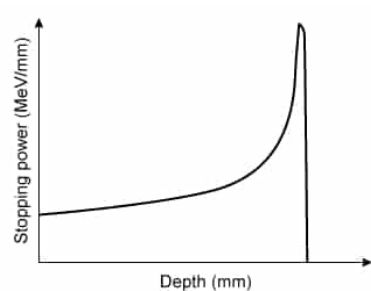


Figure 2 Stopping power vs depth in tissue

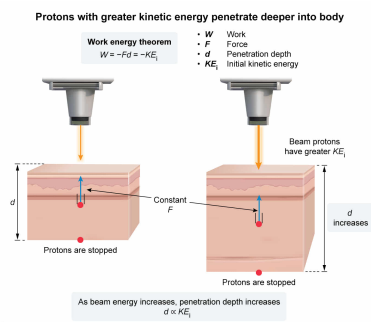
Because most of the protons' energy loss is concentrated near the peak of the curve, the radiation can be targeted directly onto the tumor at a specific depth. (Note: $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$)

As the kinetic energy of the protons in the beam is increased, what happens to the average depth into the body that the protons penetrate? (Note: Assume the protons move in a straight line and slow down at a constant rate.)

- ☐ A. The depth decreases. [10%]
- ☒ B. The depth increases. [57%]
- ☐ C. The depth remains constant. [9%]
- ☐ D. The depth increases, then decreases. [23%]

Correct
57% answered correctly

Explanation:



According to the **work-energy theorem**, the **work** W done on an object equals the change in its **kinetic energy** ΔKE :

$$W = \Delta KE$$

Furthermore, when a constant force is exerted on an object, W equals the product of the force **component** $F_{\parallel}$ along the direction of the

object's motion and the distance d that the object moves as the force is exerted. Hence:

$$W = F_{\parallel} d = \Delta KE$$

In this question, the protons in the beam have an initial kinetic energy KE_i but their final kinetic energy KE_f is zero because they come to rest in the tissue. Therefore:

$$\Delta KE = KE_f - KE_i = -KE_i$$

As such, the work-energy theorem implies that the work done on the protons is the negative of KE_i :

$$W = -KE_i$$

Because the protons travel approximately in a straight line and lose speed at a constant rate, the force F that they experience is approximately constant and directed opposite to their displacement. Consequently, $F_{\parallel} = -F$ and thus:

$$W = -Fd = -KE_i$$

Solving this equation for d (ie, the penetration depth) shows that d is **directly proportional** to KE_i :

$$d = \frac{KE_i}{F}$$

$$d \propto KE_i$$

Therefore, as the beam energy is increased, the depth of penetration also increases.

(Choices A, C, and D) As the beam energy increases, a decreasing, constant, and variable penetration depth is inconsistent with the work-energy theorem.

Educational objective:

The work-energy theorem implies that particles with greater kinetic energy penetrate deeper into a material such as body tissue.

Time spent: 0 sec
Subject:
Physics

QID: 403118
System:
Mechanics and Energy

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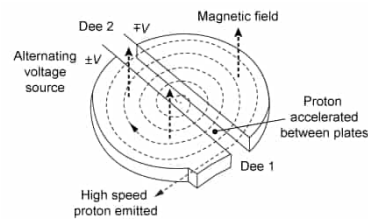


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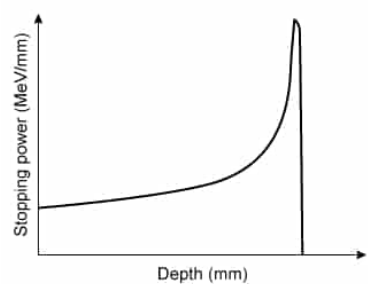


Figure 2 Stopping power vs depth in tissue

Because most of the protons' energy loss is concentrated near the peak of the curve, the radiation can be targeted directly onto the tumor at a specific depth. (Note: $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$)

The radiation dose received by an object is measured in units of grays (Gy), where $1\text{ Gy} = 1\text{ J/kg}$. A technician uses a proton beam that delivers energy at a rate of $1 \times 10^{16}\text{ eV/s}$ to treat a 50 g region of tissue. If the beam runs for 1 min, what is the dose received by the tissue?

- ☐ A. 0.03 Gy [15%]
- ☐ B. 0.2 Gy [20%]
- ☒ C. 2 Gy [59%]
- ☐ D. 100 Gy [4%]

Correct
58% answered correctly

Explanation:

Computing radiation dose from beam power and time interval

$$P = \frac{E}{\Delta t}$$

$$\text{Dose} = \frac{E}{m}$$

- P Power
- E Energy delivered
- Δt Time interval
- E Radiation dose
- m Mass

Beam $P = 1 \times 10^{16}\text{ eV/s}$
($1.6 \times 10^{-3}\text{ J/s}$)

Tumor
($m = 50\text{ g of tissue}$)

Beam runs for 1 minute

$$\text{Dose} = \frac{P\Delta t}{m} = \frac{(1.6 \times 10^{-3}\text{ J/s})(60\text{ s})}{0.050\text{ kg}} = 1.9\text{ J/kg} \approx 2\text{ Gy}$$

Power P is defined as the rate at which energy E is consumed, expended, or transferred over an interval Δt :

$$P = \frac{E}{\Delta t}$$

In this question, the proton beam transfers energy to the tissue at a rate equal to 1×10^{16} eV/s, which is the power of the beam:

$$P = 1 \times 10^{16} \frac{\text{eV}}{\text{s}}$$

The total amount of energy delivered to the tissue over Δt can be determined by solving the power equation for E :

$$E = P \cdot \Delta t$$

Inserting the beam's power and a Δt of 1 min yields:

$$\begin{aligned} E &= \left(1 \times 10^{16} \frac{\text{eV}}{\text{s}}\right) \cdot (1 \text{ min}) \\ E &= \left(1 \times 10^{16} \frac{\text{eV}}{\text{s}}\right) \cdot \left(\frac{60 \text{ s}}{1 \text{ min}}\right) \cdot (1 \text{ min}) \\ E &= 6 \times 10^{17} \text{ eV} \end{aligned}$$

Converting eV to joules yields:

$$\begin{aligned} E &= 6 \times 10^{17} \text{ eV} \cdot \left(1.6 \times 10^{-19} \frac{\text{J}}{\text{eV}}\right) \\ E &= 9.6 \times 10^{-2} \text{ J} \end{aligned}$$

The radiation dose is defined as the amount of E per unit mass m :

$$\text{Dose} = \frac{E}{m}$$

Inserting the value of E and the tissue mass of 50 g gives:

$$\text{Dose} = \frac{9.6 \times 10^{-2} \text{ J}}{50 \text{ g}}$$

Converting to units of grays (J/kg) yields:

$$\begin{aligned} \text{Dose} &= \frac{9.6 \times 10^{-2} \text{ J}}{50 \text{ g}} \cdot \left(\frac{1,000 \text{ g}}{1 \text{ kg}}\right) \\ \text{Dose} &\approx 2 \frac{\text{J}}{\text{kg}} = 2 \text{ Gy} \end{aligned}$$

(Choice A) 0.03 Gy follows from neglecting to multiply the power by 60 s when finding the energy delivered.

(Choice B) 0.2 Gy follows from converting 50 g to 0.5 kg instead of 0.05 kg in the formula for the dose.

(Choice D) 100 Gy results from incorrectly multiplying the power by 3,600 s (1 hr) instead of 60 s (1 min).

Educational objective:

Power is the rate of energy transferred per unit time. A radiation dose can be determined by multiplying power by the time interval and then dividing by the target mass.

Time spent: 0 sec
Subject:
Physics

QID: 403119
System:
Mechanics and Energy